Numerical Decidability for Sound SciML Metamorphic Testing

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Abstract

Scientific machine-learning (SciML) surrogates are hard to test because exact outputs are unavailable for transformed inputs. Metamorphic testing can turn physical relations into oracle-free tests, but a candidate relation is not automatically a sound verdict: a failure may reflect a surrogate inconsistency, an invalid relation application, or a numerical artifact of the measurement operator. This paper treats numerical decidability as a soundness precondition for SciML metamorphic testing. The workflow proposes a candidate relation, checks whether it is physically valid and numerically measurable, executes only the admitted checks, assigns a typed verdict, and restricts paper claims to that verdict. For P1 discrete divergence on shape-regular triangular meshes, we give a local measurement-floor bound and instantiate it on the cylinder-flow mesh, with a second Delaunay topology used as bounded stability evidence. The gate changes verdicts rather than only annotating them: it defers absolute conservation when the floor dominates, rejects an inadmissible airfoil continuity relation on physical-basis grounds, and keeps node permutation as an implementation sanity check. Bounded MeshGraphNets, PointMLP, PINN, FNO, airfoil, and cross-program executions show that de-

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tector coverage follows the admitted relation set. The evidence supports an auditable admissibility workflow, not general SciML reliability, baseline superiority, arbitrary-mesh soundness, or real-world defect-detection rates.

Keywords: Metamorphic testing, metamorphic relation identification, oracle problem, scientific machine learning, neural surrogates, MeshGraphNets, domain-validity rubric

1. Introduction

Scientific machine-learning (SciML) surrogates are now used as software components for approximating expensive simulations [1, 2, 3]. Mesh-based neural simulators such as MeshGraphNets (MGN) predict physical fields on irregular meshes [4], but they also inherit a classical testing difficulty: for many candidate inputs there is no cheap exact output against which the program can be checked, the test oracle problem [5]. Held-out rollout error is therefore necessary, but it does not by itself tell a tester whether the software respects relations that should hold under physically meaningful transformations.

Metamorphic testing (MT) addresses the oracle problem by checking relations among multiple executions rather than a single expected output [6, 7]. Conservation laws, symmetry relations, nondimensional similarity, and continuity conditions can all suggest candidate metamorphic relations (MRs) for scientific software [8, 9, 10]. The testing problem considered in this paper begins after such candidates have been proposed. A physics-derived relation is not automatically a valid software test: mirror symmetry may depend on geometry and boundary labels; a divergence relation may depend on a discrete operator, mesh weights, and boundary treatment; a transformed case may leave the relation's domain before it reveals any system-under-test (SUT) inconsistency.

For SciML surrogate software, a failed MR check may indicate SUT inconsistency, invalid relation use, incompatible output mapping, or an unresolvable scoring floor. Treating all outcomes as ordinary pass/fail MT verdicts makes the evidence hard to audit and easy to overclaim.

Here, a candidate relation becomes an oracle-free test only when it is valid for the SUT and measurable for the deployed discretization. The paper's contribution is a bounded workflow for making that decision before a failed check is reported as software evidence. The paper does not replace residual, uncertainty, accuracy, or trust-region diagnostics. A downgraded relation may still be OOD stress evidence, but not a SUT fault verdict. The evidence supports bounded SciML V&V utility rather than general reliability or real-world defect-detection rates.

Reader map.. The method uses five concepts. A *candidate relation* is a proposed property that the SUT should preserve under a controlled change. A *validity gate* asks whether that relation is meaningful for this SUT and transformed case. *Numerical decidability* is the measurement part of the gate: the metric and tolerance must be large enough to distinguish a real relation violation from the measuring operator's floor. An *executable check* is the runnable source/follow-up test built from an admitted relation. A *typed verdict* records whether the check passes, fails inside the relation domain, is outside the relation domain, is numerically unresolved, or is inconclusive. In one sentence, the paper screens a proposed relation, executes it only when valid and measurable, records a typed verdict, and restricts claims to that verdict. All other terms in the paper are implementation details or evidence records built from these five concepts.

Two principles organize the method. First, a relation is admitted only when its validity conditions hold for the SUT and its tolerance dominates the measuring operator's intrinsic floor. Second, relation outcomes are interpreted along relation-violation and domain-violation axes, separating SUT inconsistency from out-of-relation-domain use, numerical-floor deferral, and inconclusive evidence. The main empirical setting is MeshGraphNets-family cylinder flow, with the same predicate exercised on a second CFD task and additional PDE families; these are bounded case studies, not a completed applicability map.

Contributions. The paper contributes a numerical-decidability gate for SciML metamorphic testing: a relation is admitted only when its verdict tolerance dominates the intrinsic floor of the measurement operator. The contribution is methodological, with domain-inadmissibility verdicts and seeded-fault stress tests reported as stress evidence rather than validated localization models.

Numerical-decidability criterion. For P1 constant-per-cell divergence on shape-regular triangular meshes and C^2 divergence-free reference fields, the operator floor is bounded locally by a mesh-shape constant times the Hessian scale and cell diameter. For the deployed cylinder-flow setting, this specializes to a tighter closed-form structured-mesh predictor that matches the measured floor to within 0.5%, a rigorous a-priori bound that dominates it, and observed topology stability across a structured mesh and an unstructured Delaunay mesh (Sec. 5.5); arbitrary degenerate or operator-mismatched meshes remain outside scope.

Executable audit mechanism. Each admitted relation is represented as a structured MR card plus executable runner that records the transformation, mapping, metric, tolerance, exclusion rule, and ledger fields needed for independent inspection.

Typed relation-level verdicts. The verdict scheme is the gate's interpretation layer. It separates relation violation from domain violation, instantiating for physics-governed SciML the constraint-architecture pattern behind bug-versus-inapplicability reasoning [11, 12] and MetaTrimmer [13]. The added element is a typed classification of physical, geometric, boundary-condition, mapping, and operator-admissibility limits.

Bounded necessity evidence. The case studies evaluate what the criterion changes beyond rollout accuracy, generic MR generation, and LLM-assisted MR elicitation, using cylinder-flow runs plus airfoil, PINN, FNO, periodic-advection, RealPDEBench, a five-unit external issue/PR/commit-linked semantic witness corpus, and cross-program checks. The evidence is complementarity and gate value, not baseline superiority or a general fault-detection-rate claim.

Paper organization. Section 2 positions the work against oracle-free testing, physics-based MT, and SciML diagnostics. Section 3 defines the gate, MR-card asset, and verdict semantics. Section 4 states the research questions, subjects, metrics, baselines, and analysis plan. Section 5 reports the empirical evidence and claim boundaries, followed by threats to validity, future work, and conclusions.

2. Related Work

2.1. Oracle-free testing and MR candidate sources

Mesh-based neural simulators learn dynamics on graph or mesh representations of physical systems. MeshGraphNets encodes mesh nodes and edges, propagates information by message passing, and predicts future physical fields through autoregressive rollout [4]. In this paper these models are treated as software SUTs. The testing question is whether a rollout is accurate on held-out trajectories and whether the SUT preserves declared relations under controlled transformations and explicit tolerances.

Metamorphic testing (MT) addresses programs lacking a per-input oracle [6, 7, 14]. Rather than checking each output against an expected value, MT checks a relation among source and follow-up executions. This framing has been applied to scientific software, PDE programs, simulations, and machine-learning (ML) systems [15, 16, 8, 10], including recent applications to tabular ML such as credit scoring [17]. For SciML surrogates, however, an MR is not merely an input perturbation: its validity depends on governing equations, boundary conditions, mesh representation, output mapping, and numerical tolerance. This makes MR admissibility part of the oracle construction problem itself.

Testing scientific software without a reliable oracle is a long-standing problem, and metamorphic testing is the most developed response in the surveyed literature [18]. MRs can be elicited from monotonicity, conservation, scaling, and domain-specific expectations [8, 9, 10, 19]. Predictive MR identification has been pursued with graph-kernel learning over scientific source code [20], by separating input and output patterns and by image-based output-pattern recognition for numerical solver codes [21, 22], and with large language models that discover metamorphic-relation variables for scientific software [23]. Symmetry-based MRs paired with statistical violation criteria have also been applied to deep trajectory predictors [24, 25]. Data-driven search can discover candidate transformations in complex physical software [26], and design assumptions can be encoded as MRs for cyber-physical-system (CPS) testing under control–plant invariants [27].

Separate work has made MR specification and applicability more explicit. General MT tooling has explored machine-readable MR representations, constraint definition, test-data-driven MR selection, and general MR specification languages [12, 28, 29]. These systems strengthen MR engineering in the large, but they do not by themselves decide whether a physics-derived relation is numerically measurable, in-domain for a transformed CFD case, or interpretable as a SciML V&V verdict.

These approaches can supply candidate sources, but candidate identification is not admissible SciML test construction. MR identification, including our own prior physics-derived MR construction for numerical solver and nuclear-power software [30, 21], is established work; this paper takes candidate MRs as input and contributes the admissibility gate rather than a new identification method. Fault-oriented MR work ranks relations by observed detection power [31, 32]; the present paper uses the admitted relation set later to interpret bounded coverage and blind spots, but treats that as a consequence of admissibility rather than as the primary novelty.

2.2. Physics-based MT, tolerance, and applicability

Three contemporary strands are closest to this paper. Yang and Chui [33] and Reichert et al. [34] applied physics-derived MRs to hydrologic ML or conceptual models, showing that learned-model accuracy and relation consistency can diverge and that calibration or forcing regimes affect interpretation. Eniser et al. [35] introduced *relaxations*, numerical tolerances embedded in MR oracles, for stochastic reinforcement-learning (RL) policy testing. Duque-Torres et al. [11, 12, 28] and MetaTrimmer [13] address bug-versus-inapplicability reasoning with explicit MR constraints and automated constraint derivation.

The present work combines these ideas in a SciML-specific V&V workflow. From physics-based MR testing it takes relation-level checks; from relaxations it takes the need for calibrated tolerances; from violation-attribution work it takes the need to avoid treating inapplicability as a SUT failure. What none of these strands supplies is the central condition used here: admissibility made contingent on numerical decidability. Relaxations attach a tolerance to an oracle, and constraint- or attribution-based work decides whether a relation applies to an input, but neither requires the verdict tolerance to dominate the measuring operator's intrinsic error floor, a property of the discretization fixed before any fault or SUT is considered. The floor gate, not the recording discipline, is what is genuinely new here. The admissibility decision, tolerance rationale, executable asset, and typed verdict are then recorded together and tied to evidence artifacts. Table 1 positions the paper against the closest-prior capabilities.

Table 1: Closest-prior capability matrix.

Approach	What it supplies	What remains missing for the evaluation objective	What this paper adds
Scientific-software MR identification	Candidate relations from domain rules, source-code learning, LLM variables, and data-driven search	Candidate validity, numerical decidability, executable asset format, and typed SciML verdicts	Treats these methods as candidate sources, not validity authorities
Reichert et al.	Physics-derived MRs and informal filtering in a learned hydrologic surrogate	Explicit admissibility predicate, operator-floor tolerance, reusable MR cards, and typed verdicts	Formalizes the candidate-to-verdict path for mesh-based SciML surrogates
Eniser et al.	Relaxed MR oracles with numerical thresholds	Measurement-operator floor and relation-domain interpretation for deterministic SciML surrogates	Grounds tolerance in the scoring operator's own numerical floor
Duque-Torres / MetaTrimmer	Binary bug-versus-inapplicability separation with explicit MR constraints and automated constraint derivation	A typed inadmissibility verdict and operator-floor admissibility for physics-governed SciML	Replaces the binary skip/proceed pre-filter with a two-axis typed verdict tied to physical, geometric, boundary, and operator limits
Formal MR specification and constraint tooling	Machine-readable MR descriptions, constraint phases, and supporting systems	Physics-specific admissibility dimensions, measurement floors, and executable evidence ledgers for SciML	Uses specification ideas but binds each verdict to relation-domain and numerical-decidability evidence
This paper	Physics, expert, and LLM-assisted candidates plus artifact-backed execution	Bounded case-study evidence, not general SciML reliability or baseline superiority	Constructs MR cards, runners, ledgers, and typed verdicts under a four-condition gate

2.3. SciML diagnostics and positioning

SciML reliability research has developed complementary tools: physics residuals, uncertainty quantification (UQ), conformal prediction, certified error bounds, and failure-mode analysis [1, 2, 3, 36, 37]. PINN training can exhibit gradient-flow pathologies that bias trained models away from the governing residual [38, 36], sharpening the need for checks that do not assume a converged surrogate. These diagnostics are not competitors to MT. A residual, equivariance error, or uncertainty score becomes an executable MR only when paired with a source case, follow-up transformation, tolerance rule, exclusion rule, and verdict interpretation. The proposed workflow uses such diagnostics as possible measurements inside a multi-execution oracle-free asset.

Hybrid ML-solver frameworks use residual thresholds or error estimates to decide when a learned component should be trusted [39, 40]. Such systems operationalize runtime trust regions, whereas this study acts offline, applying physically derived transformations before deployment to estimate where relation violations occur.

The distinction from residual- and uncertainty-based trust estimation is deliberate. UQ, conformal prediction, residual-threshold trust regions, and physical-consistency diagnostics that score a physically derived residual on observed outputs [41] locate unreliable behavior from observed inputs. The present method acts in relation space: it applies a controlled transformation and reports which necessary relation breaks under it, indexed by that transformation. The claim is complementarity, not superiority: relation-indexed evidence answers a different question from scalar accuracy, residual magnitude, or uncertainty.

Metamorphic testing, MR identification, scientific-software MT, residual diagnostics, uncertainty quantification, LLM candidate generation, and the NOETHER two-layer MetaPattern model are established or emerging sources of testing ideas. The contribution here is narrower: SciML MR use should be treated as a domain-validity and asset-construction problem. A candidate relation becomes a useful oracle-free V&V asset only after its physical basis, transformation preconditions, output mapping, tolerance, exclusion rule, executable artifact, and relation-level verdict are recorded. Recent work derives or refines domain-specific MRs for scientific programs directly from their numerical models [42, 43]; the wedge here is the learned surrogate evaluated out of distribution and the requirement that a relation's tolerance dominate the measuring operator's numerical floor, rather than testing a numerical solver against derived relations. The missing combination is thus a validity-gated, executable, and auditable workflow for physics-governed SciML surrogate software (Table 1).

3. Method

3.1. Overview

The method turns a physics-derived candidate relation into an auditable V&V asset only after three checks are made explicit: validity gating, executable-check construction, and verdict interpretation. These design roles are evaluated by the research questions introduced in Section 4.1. Figure 1 summarizes the workflow. Candidate generation is deliberately separated from validity judgment: expert reasoning, physical laws, the NOETHER two-layer MetaPattern model, and LLM-assisted lists may propose relations, but none certifies a relation as valid for a specific SUT.

Formal spine. Let $r = (b, T, M, m, \tau, P)$ denote a candidate relation, where b is its basis, T the source-to-follow-up transformation, M the output mapping, m the metric, τ the tolerance, and P the preconditions. For SUT s and source case x, the gate returns $G(r, s, x) \in \{\text{admit}, \text{reject}, \text{stress}, \text{defer}\}$. If admitted or retained as stress evidence, execution returns $E(r, s, x) = (y, y', z)$, where y and y' are source and follow-up outputs and z is the measured relation evidence. The verdict map $V(G, z)$ assigns the typed verdict. Only an admitted relation with a fail verdict inside the relation domain can support a SUT-inconsistency claim.

Terminology and verdict types. A verdict is read on two axes, relation-violation and domain-violation magnitude (detailed in Sec. 3.5). A candidate relation is accordingly *admitted* (a large in-domain violation reads as SUT inconsistency), *rejected* on physical grounds, *deferred* when no tolerance dominates the operator floor, *out-of-relation-domain*, or *inconclusive*. Four MR families recur in the case studies: node-permutation equivariance, mirror-y reflection symmetry, discrete-divergence continuity/conservation, and exact mirror-y on a constructed symmetric mesh. As a running example, a mirror-y candidate is first checked against geometry and boundary labels; on an asymmetric mesh it becomes OOD stress rather than a fault oracle, while on a constructed symmetric mesh it remains an admitted equivariance test.

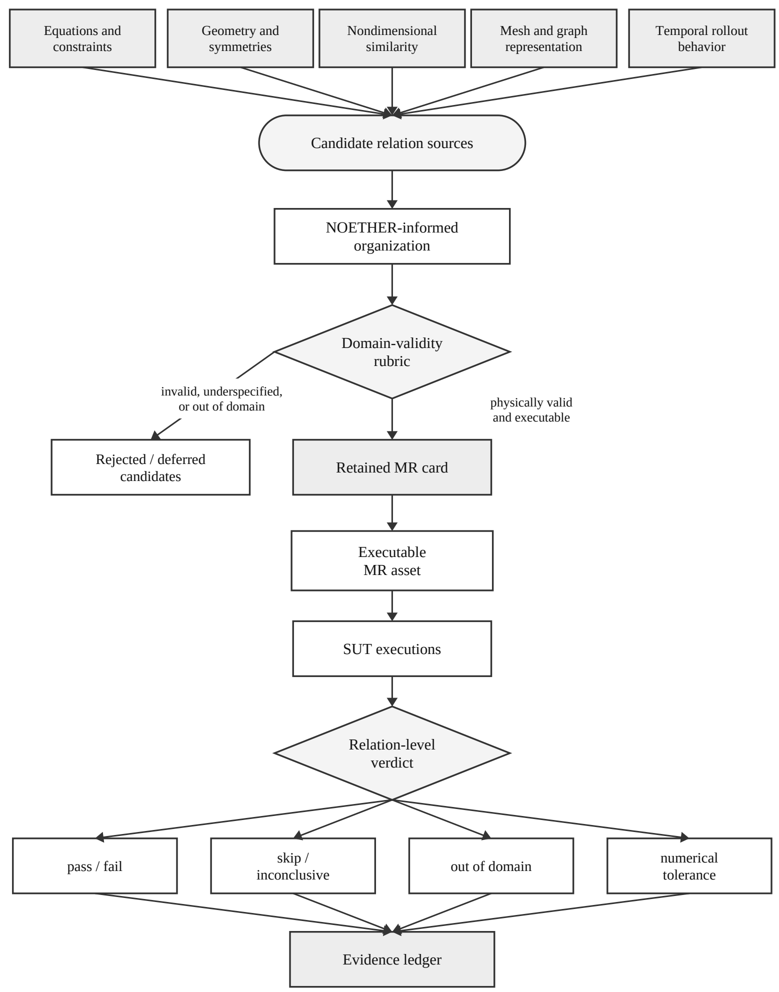

Figure 1: Validity-gated V&V workflow.

3.2. Candidate relation sources: a two-layer MetaPattern model

We organize candidate relations with the two-layer MetaPattern model of NOETHER [44], a same-author preprint, which grounds each relation in an explicit algebraic generating structure, superseding the earlier descriptive three-level physical/computational/code classification [45]. This two-layer organization is an optional scaffold for generating and naming candidate relations; the central decision point is the admissibility gate of Section 3.3, which applies however a candidate was sourced.

Layer 1 (MetaPattern). A *MetaPattern* is the equivalence class of MRs generated, through TRANSLATE, from one minimal algebraic structure of the program-induced operator algebra $\mathcal{A}_P$. The five MetaPatterns are generated by a group action (G), partial order ($O_{\leq}$), self-adjoint operator (T^*), time-reversal involution ($\mathcal{T}^*_{\mathrm{rev}}$), or parametrized limit ($\mathcal{L}^*$); their set is $\mathbb{M}(\mathcal{A}_P)$. Conservation is therefore the group-action member m_{inv}.

Layer 2 (MR family). An *MR family* instantiates one MetaPattern under a fixed transformation template and mode. Families use the form $\mathsf{f}_{\mathrm{parent.child}}$; legacy letters a–j remain cross-reference anchors. The map is one-to-many: $G \to \{\mathsf{f}_{\mathrm{inv.eqv}}, \mathsf{f}_{\mathrm{inv.con}}\}$, $T^* \to \{\mathsf{f}_{\mathrm{adj.self}}, \mathsf{f}_{\mathrm{adj.dual}}\}$, $\mathcal{T}^*_{\mathrm{rev}} \to \{\mathsf{f}_{\mathrm{rev.traj}}\}$, $O_{\leq} \to \{\mathsf{f}_{\mathrm{mono.stat}}, \mathsf{f}_{\mathrm{mono.shape}}\}$, and $\mathcal{L}^* \to \{\mathsf{f}_{\mathrm{conv.lim}}, \mathsf{f}_{\mathrm{conv.rate}}, \mathsf{f}_{\mathrm{conv.repr}}\}$.

For the cylinder-flow surrogates studied here, the instantiated families are $\mathsf{f}_{\mathrm{inv.eqv}}$ (equivariance, a), mirror-y reflection; $\mathsf{f}_{\mathrm{inv.con}}$ (conservation, b), discrete-divergence continuity and, on the airfoil task, compressible mass balance; $\mathsf{f}_{\mathrm{conv.repr}}$ (representation-invariance, j), node-permutation and encoding contracts; and $\mathsf{f}_{\mathrm{conv.lim}}$ and $\mathsf{f}_{\mathrm{conv.rate}}$ (convergence, accuracy-order, h–i), the operator-floor mesh-refinement relation. The $\mathsf{f}_{\mathrm{mono.stat}}$ (static-order, f) Reynolds–Strouhal monotonicity is a candidate not exercised here, $\mathsf{f}_{\mathrm{rev.traj}}$ (trajectory-reversal, e) is empty by construction for viscous dissipative flow, and the self-adjoint families $\mathsf{f}_{\mathrm{adj.self}}$ and $\mathsf{f}_{\mathrm{adj.dual}}$ (c–d) appear only in the cross-family PINN and FNO extensions. The same relation may be admissible as one family yet inadmissible once instantiated for a given SUT: mirror symmetry holds on a symmetric domain but not on an asymmetric mesh with incompatible boundary labels, and conservation is a physical expectation but not an executable verdict when the discrete measuring operator has an unresolved floor. The admissibility gate records where each relation survives.

3.3. Admissibility gate

A candidate relation is an *admissible MR* only when four conditions hold together. First, it must have a physical or software basis. Second, its transformation preconditions must hold for the source and follow-up cases. Third, boundary conditions and output mappings must remain compatible after transformation. Fourth, the verdict must be numerically decidable: the tolerance must dominate the intrinsic error floor of the measuring operator. None of these conditions is optional. Table 2 records the gate, the MR-card fields, and the verdict semantics.

Table 2: Admissibility gate and verdict semantics.

Method element	**Required record**	**Allowed interpretation**	**Design role**
Basis	Equation, boundary condition, nondimensional law, representation property, numerical assumption, or implementation contract	Relation may be considered only for SUTs where this basis applies	Gate
Preconditions	Source-case constraints, follow-up transformation, preserved quantities, and exclusion rule	Missing preconditions produce skip or out-of-relation-domain, not SUT failure	Gate
Mapping compatibility	Boundary-label compatibility and output mapping from follow-up output back to source coordinates/fields	Incompatible mapping rejects or downgrades the candidate	Gate
Numerical decidability	Metric, tolerance, tolerance provenance, and measurement-floor evidence	A verdict is meaningful only when tolerance dominates the operator floor	Gate
MR card	Identifier, source category, transformation, mapping, metric, tolerance, exclusions, expected verdicts, and artifact schema	The admitted relation becomes a reusable executable V&V asset	Asset
Evidence ledger	Source/follow-up run ids, metric outputs, floor evidence, exclusions, logs, and verdict	The reported claim is auditable from artifacts rather than prose alone	Asset
Verdict semantics	Pass, fail, skip, out-of-relation-domain, numerical-tolerance issue, or inconclusive	Only fail inside the valid domain may be read as SUT inconsistency	Verdict

The gate produces four design-time decisions: admit as an executable MR; downgrade to OOD-stress when the relation is physically informative but outside its exact domain; reject when basis, preconditions, or mapping fail; or defer when missing numerical evidence prevents a verdict. For example, node permutation can be admitted as a representation MR when the adapter and inverse mapping are deterministic; mirror-y is conditional on compatible geometry, boundary labels, and vector-component mapping; absolute discrete divergence is deferred when the reference field and measuring operator floor cannot support an absolute verdict.

3.4. MR card and executable asset format

A retained MR is represented as a structured MR card recording the fields listed in Table 2: identifier, source category, physical or software basis, source-case schema, follow-up transformation, preconditions, boundary-condition compatibility, output mapping, metric, tolerance and provenance, expected verdict classes, exclusion rules, artifact schema, and interpretation limits. The card is the reusable design record; the executable runner is the operational artifact generated from it.

Each executable asset includes transformation code, runner configuration, metric computation, verdict logic, and artifact recording. The ledger is part of the method rather than only replication packaging: every relation-level claim must be traceable to the source run, follow-up run, mapping, metric output, floor evidence, exclusion decision, and verdict.

3.5. Relation-level verdicts

An MR execution can produce: **pass** (relation holds within tolerance); **fail** (violated within the valid relation domain); **skip** (required precondition missing); **out-of-relation-domain** (the transformed case is outside the relation domain); **numerical-tolerance issue** (the measured effect is dominated by threshold or resolution uncertainty); or **inconclusive** (insufficient artifact).

These verdicts form the two-dimensional space in Figure 2. One axis is relation-violation magnitude, typically expressed as a violation-to-floor or violation-to-tolerance ratio. The other is domain-violation magnitude: how far the transformed case lies outside the relation's validity domain. Low domain violation with high relation violation is the region that may be read as SUT inconsistency; high domain violation is out-of-relation-domain; a violation inside the error floor is a numerical-tolerance issue. This decomposition defines the method's verdict-interpretation role, which is evaluated explicitly in Section 4.1.

In this study, the domain-violation coordinate is operationalized per relation from committed measurements. For mirror-y, $D = m/(m+1)$, where m is the worst reflected-node placement error in median-edge-length units. The synthetic symmetric mesh scores $D \approx 0$, whereas the real asymmetric evaluation mesh scores $D = 0.51$. Other MR families use their own precondition measurements. These D values denote structural placement within each relation's own validity domain: D is a per-relation normalized coordinate, not a cross-relation calibrated metric, and the values cannot be averaged or ranked across MR families; read per relation, not as a calibrated boundary measurement. A cross-relation calibration would need to normalize each family by its own admissibility boundary, mapping every gate threshold to a common $D_{\mathrm{cal}} = 1$; that calibration is future work.

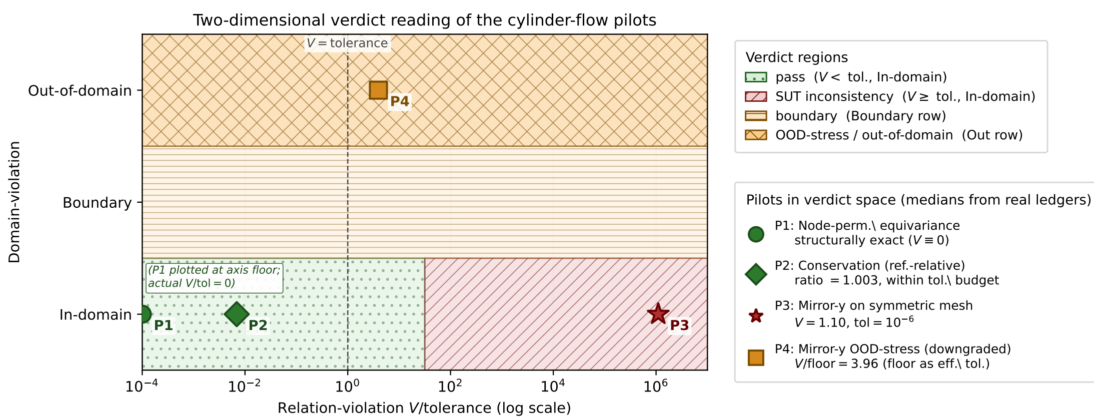

Figure 2: Two-dimensional relation and domain verdict space.

Retained MRs are interpreted through their two-layer position (Sec. 3.2): a relation's MetaPattern and MR family fix which algebraic invariant it measures, and therefore which faults it can and cannot surface. This is an interpretation protocol, not a validated localization model; the seeded-fault results later are a bounded stress test of it.

Practitioner checklist and adoption boundary.. For a new SUT: (1) state the relation and source; (2) record source/follow-up preconditions; (3) check boundary labels and output mapping; (4) bound or calibrate the operator floor; (5) compare verdict tolerance with that floor; and (6) issue a typed verdict and ledger the run. The method does not remove domain expertise or automate MR discovery. Adoption costs are relation-specific physics review, mapping code, floor evidence, and bookkeeping; when any item is missing, the outcome is reject, defer, or OOD-stress, without a fault claim.

4. Experimental Design

4.1. Research questions

The experimental design is organized around the following objective: to evaluate whether physics-derived MRs for SciML surrogate software can be transformed into auditable, executable, and interpretable oracle-free V&V assets that distinguish genuine SUT inconsistency from relation invalidity and numerical measurement limits.

This objective is decomposed into four research questions:

RQ1 – Screen. Which validity conditions screen a physics-derived candidate MR before it is used as an oracle-free test for a specific SciML surrogate SUT?

RQ2 – Build. How can admitted relations build reusable executable checks with explicit transformations, mappings, metrics, tolerances, exclusions, and evidence records?

RQ3 – Interpret. How can typed verdicts interpret relation-level outcomes so that SUT inconsistency is separated from out-of-relation-domain application, numerical tolerance limits, and inconclusive evidence?

RQ4 – Evaluate. In bounded SciML surrogate case studies, what evidence can evaluate the validity-gated workflow beyond rollout accuracy, generic MR generation, and LLM-assisted candidate elicitation?

4.2. Experimental subjects

Subjects are selected for evidence roles, not statistical representativeness: a primary end-to-end SciML surrogate setting, a changed physical task or architecture, a cross-family PDE/neural-operator setting, or a bounded defect/coverage witness. We do not sample by repository popularity, LOC, or issue/changelog-mined defects; that would be a separate real-defect study. The evidence supports bounded V&V utility and coverage interpretation, not population-wide defect-detection effectiveness.

The executable inventory comprises two CFD tasks, same-task MGN variants, PointMLP, PhysicsNeMo, PINN/FNO rosters over two PDEs, one periodic-advection workflow, and five external CPU-only replays. Fault evidence comprises 10 canonical and two adversarial MGN mutants, 48 PINN/FNO probes, a PointMLP 20-fault catalogue, and the sibling's 70 witness rows plus 20 true-fault-class rows. These are scope denominators, not sampling frames for a defect-detection rate.

Table 3: Subject selection and evidence scope.

Evidence block	**Subjects and evidence role**	**Boundary**
Cylinder-flow mesh surrogates	Primary executable workflow plus same-task implementation checks: MGN K=6 roster over three trajectories, S4/S5 variants, PointMLP, and PhysicsNeMo `MeshGraphNet`; seeded-fault stress uses 10 canonical MGN mutants, two adversarial mutants, and PointMLP 20-fault catalogue	Same task and bounded rosters; no real-defect rate or LOC-based representativeness
Airfoil mesh surrogate	Changed-physics gate discrimination on official DeepMind airfoil with PhysicsNeMo `MeshGraphNet`, K=6 checkpoints over 40 test trajectories	Typed inadmissibility evidence, not SOTA airfoil accuracy or official-checkpoint evidence
PINN/FNO and periodic PDE workflows	Cross-family PINN/FNO Burgers/heat predicate checks plus independent periodic-advection primary workflow with six periodic-convolution SUTs and 60 relation cells per admitted MR	Synthetic/generated PDE evidence; not production CFD, real-defect, reliability, or broad neural-operator claims
External witnesses	RealPDEBench foil preflight witness, five-unit external issue/PR/commit-linked semantic witness corpus (including the DeepXDE PeriodicBC issue-linked witness, NeuralOperator, PhiFlow/PhiML, and JAX-CFD), and Minimum-MR-SubSet audit/replays	Triangulates outside the self-made cylinder setting; not production-CFD validation, trained-SUT correctness, or a representative defect-rate corpus

Independent evidence units include cylinder-flow trajectories, same-task implementations, airfoil, PINN/FNO families, periodic advection, and seeded/external witness rows. Effective-N and inference permissions are in the supplement; repeated cells remain descriptive evidence, not population samples.

Primary evidence comes from two bounded full-chain workflows: MeshGraphNets-family cylinder flow (trained checkpoint, K=6 roster, S4/S5, PointMLP, and PhysicsNeMo `MeshGraphNet`) and independent periodic advection (six NumPy periodic-convolution SUTs, ten held-out fields per SUT, two admitted MRs, one rejected candidate). The unit is a relation-level cell: MR card, SUT/checkpoint, source, follow-up, metric, and verdict.

Supporting evidence checks whether the same admissibility predicate applies outside the primary mesh-surrogate setting: airfoil gate discrimination, a fifteen-seed PINN roster (K=15), and trained FNO-2D checkpoints over 2D Burgers and heat data.

Secondary evidence includes LLM-simulated expert MR design, generic MR-generation scope contrasts, LLM-assisted candidate generation, rollout-accuracy diagnostics, a production-adjacent RealPDEBench preflight, a five-unit external issue/PR/commit-linked semantic witness corpus, and external witness reruns from the read-only Minimum-MR-SubSet repository.

Stress-test evidence uses seeded-fault and cross-program checks to interpret where the admitted MR set can and cannot observe faults.

4.3. Evaluation metrics

Primary metrics are admissibility decision, executable MR rate, verdict distribution, violation magnitude relative to tolerance or floor, exclusion rate, and relation-level interpretation output. Secondary metrics include inter-rater agreement for candidate screening, flakiness rate, complementarity with rollout accuracy, and stress-test detection by seeded-fault class. These metrics are interpreted at the relation-cell level rather than as general reliability or defect-detection rates.

4.4. Experimental method

The experiments instantiate the candidate-to-asset workflow of Section 3. The relations evaluated are the cylinder-flow MR families of Section 3.2: equivariance ($\mathsf{f}_{\mathrm{inv.eqv}}$) and conservation ($\mathsf{f}_{\mathrm{inv.con}}$) under m_{inv}, representation-invariance ($\mathsf{f}_{\mathrm{conv.repr}}$) and the convergence/accuracy-order pair ($\mathsf{f}_{\mathrm{conv.lim}}$, $\mathsf{f}_{\mathrm{conv.rate}}$) under m_{conv}; the static-order ($\mathsf{f}_{\mathrm{mono.stat}}$) and self-adjoint ($\mathsf{f}_{\mathrm{adj.self}}$, $\mathsf{f}_{\mathrm{adj.dual}}$) families are exercised only as candidate or cross-family checks, and trajectory-reversal ($\mathsf{f}_{\mathrm{rev.traj}}$) is empty by construction. For each retained relation, the workflow records an MR card, source and follow-up executions, transformation and mapping artifacts, metric outputs, floor evidence, exclusions, and a typed verdict. The design separates four evidence tiers: primary, supporting, secondary, and stress-test evidence.

4.5. Comparison baselines

Rollout accuracy, residual/UQ diagnostics, expert-designed candidates, generic MR generators, and LLM-assisted candidates contextualize RQ4. Rollout accuracy asks whether a prediction is close to observed held-out data; an MR asks whether a declared relation holds under a controlled transformation. The baseline contrast is therefore complementarity and gate value: false-alarm removal, typed triage, auditability, and executable-asset reuse, not superiority in a shared scalar score. LLM and generic MR outputs are treated as candidate sources; the admissibility gate, not the generator, decides whether a candidate becomes an executable MR.

4.6. Analysis plan and evidence mapping

Table 4 maps research questions to evaluation evidence, artifacts, units of analysis, and claim boundaries; it is a design rather than a result table.

Table 4: Evaluation design.

Item	Evaluation question	Evidence tier and unit	Artifacts	Claim boundary
Goal	Does the full workflow connect candidate, gate, executable asset, verdict, and evidence?	Primary workflow trace; MR-card cell	MR cards, runners, claim ledger	Asset construction only; no general reliability claim
RQ1	Which candidates are admissible, rejected, downgraded, or deferred?	Primary and supporting gates; candidate/MR	Rubric records, operator-floor sweep, exclusions	Physical validity is relation- and SUT-specific
RQ2	Are admitted relations executable and auditable?	Primary execution cells; source/follow-up pair	Transformations, runner configs, metric files, logs	Execution completeness, not complete automation of MR discovery
RQ3	Do verdicts separate SUT inconsistency from domain and numerical limits?	Verdict ledger; MR-by-case cell	Two-axis verdicts, metric/floor ratios, exclusion logs	Per-relation interpretation; no cross-relation calibrated domain score
RQ4	What evidence is added beyond rollout accuracy and candidate-generation comparators?	Primary, supporting, secondary, and stress-test tiers	Rollout diagnostics, LLM/generic baselines, witness reruns, seeded-fault ledgers	Complementarity and bounded utility; no baseline superiority
Impl.	What blind spots follow from the admitted MR set?	Stress-test tier; mutant/MR cell	Seeded-fault catalogue and cross-SUT removal check	Bounded coverage interpretation, not real-world detection rate

The study reports inferential statistics only where the sampling structure supports them, labeling descriptive counts otherwise. Binary detector outcomes over the unified fault catalogue carry Wilson 95% confidence intervals (CIs) per detector; continuous violation magnitudes use exact Wilcoxon signed-rank tests with a rank-based effect size (Cliff's δ), where the design supports paired comparison; roster medians carry paired bootstrap intervals; and the operator-floor sweep reports a log-log slope with a 95% CI. Repeated cells from one trajectory or one checkpoint family are reported as descriptive cell summaries, not independent population samples.

4.7. Ethics, integrity, and reproducibility

The study does not involve human subjects or personal data. SUT versions, datasets, checkpoints, scripts, and run logs are recorded when licensing permits. Failed, skipped, out-of-relation-domain, numerical-tolerance, and inconclusive outcomes remain in the ledger. LLM use is restricted to candidate generation and evidence organization, not validity judgment. No candidate MR is described as valid without a passing gate decision and MR card; no OOD violation is described as a program fault without supporting verdict evidence.

5. Results and Discussion

Evidence is reported by research question rather than chronology (Table 4). Roles keep breadth from becoming representativeness: cylinder-flow and periodic-advection are bounded primary full-chain workflows; airfoil, PINN, and FNO are supporting falsification checks; LLM/generic baselines and sibling reruns are secondary contrasts; and seeded faults stress detector blind spots. Baseline contrasts measure gate value, not method superiority.

5.1. Evidence overview

The compact claim-to-evidence map binds claims to tracked artifacts; the full table is supplementary, and `claim-ledger.yml` remains authoritative.

Table 5 is a compact verdict index; read each row as the same chain–candidate relation, gate decision, runtime verdict, and claim boundary–rather than as an independent population sample. Detailed paths and per-run artifacts are supplementary and ledgered.

Table 5: MR-card verdict map.

Evidence block	Rubric outcome	Runtime verdict	Boundary

Cylinder-flow MGN/variants	Node permutation admitted; mirror exact relation downgraded or admitted only on symmetric mesh; absolute conservation deferred	Node permutation exact; mirror OOD stress fails 180/180; exact symmetric-mesh mirror fails 18/18; reference-relative conservation passes 162/162	Primary bounded workflow; absolute conservation remains deferred; no geometry-independent or population rate
Same-task architecture checks	Same predicate applied to S4/S5, PointMLP, and PhysicsNeMo cylinder executions	S4/S5 and PointMLP reproduce typed verdict pattern; PhysicsNeMo smoke records node-permutation pass and mirror OOD stress	Implementation artifact check, not production-scale reliability
Airfoil changed-physics task	Node permutation admitted; incompressible continuity and mirror-y rejected/deferred for physical/numerical reasons	240/240 node-permutation exact; continuity and mirror verdict types change with compressible/non-zero-angle physics	Gate discrimination, not SOTA airfoil accuracy
FNO primary workflow upgrade	Translation admitted; periodic discrete-conservation admitted-with-reference-floor; Dirichlet translation rejected	24/24 translation passes, 24/24 conservation failures, and 6/6 rejected Dirichlet exact-MR executions	Full rubric-to-verdict FNO evidence, outside cylinder-flow and broad neural-operator claims
Periodic-advection primary workflow	Periodic translation and mass/mean conservation admitted; fixed-velocity mirror candidate rejected	60/60 translation passes, 60/60 mass-conservation passes, and 6/6 fixed-velocity mirror rejections	Independent synthetic PDE workflow; not production CFD or real-defect evidence
External defect-witness corpus	Five documented external issue/PR/commit semantic contrasts admitted at component/utility level	5/5 typed pass: DeepXDE derivative periodicity, NeuralOperator spectrum and Hermitian fixes, PhiML gradient axis order, JAX-CFD flux boundary	Stronger external-validity witness set; not representative defect sampling, trained-SUT correctness, production validation, or a defect rate

5.2. Primary cylinder-flow evidence

All pilots use one trained MeshGraphNets cylinder-flow surrogate from the read-only Minimum-MR-SubSet checkpoint. They exercise three rubric outcomes, with raw outputs, manifests, and metric ledgers mapped in the supplementary table.

The primary cylinder workflow has six readings: node permutation is an exact representation sanity check; asymmetric mirror-y is downgraded to OOD-stress and fails on 10/10 frames; absolute continuity is deferred while the reference-relative diagnostic is an *inconclusive: reference-relative non-regression guard*; rollout accuracy answers a different question; exact mirror-y fails only after a symmetric mesh makes the relation admissible; and seeded faults show MR-family-specific blind spots.

Representation MR. Node-permutation equivariance holds to machine precision (relative L2 = 0.0 at tolerance 1e-6), a pipeline sanity check rather than model-capability evidence.

Geometric MR. Exact mirror-y is out-of-relation-domain on the asymmetric mesh: reflection is non-bijective, the worst reflected-node mismatch is about one edge length, and the cylinder is off-centre by 7.2 mm. The downgraded nearest-neighbour OOD-stress probe **failed on 10 of 10 recorded eval frames** (median relative L2 0.737, median V/floor 3.96), with no skipped or inconclusive frame. This is descriptive evidence from one SUT, checkpoint, MR, and trajectory; the symmetric-mesh run tests whether the failure is only geometric.

Continuity MR. The P1 discrete-divergence operator gives non-negligible divergence even on ground truth, so absolute mass conservation is deferred. Predicted divergence stays within about 0.4–0.8% of the reference, with an interior-only check against a pure boundary artifact. Because the reference term is not decomposed into operator error, solver artifact, or non-solenoidal data, the verdict is *inconclusive: reference-relative non-regression guard*, not conservation. The 50% threshold means only "no > 1.5 regression," not "conserves mass."

Rollout-accuracy diagnostic. On the same trajectory, one-step next-state error has median relative L2 0.0216 (mean 0.044) over nine transitions. The mirror-y OOD-stress violation (median 0.737) is about 34x the median accuracy and 17x the mean. These are different relative-L2 objects, so the contrast is one-step order-of-magnitude evidence, not rollout-stability evidence.

Exact mirror-y on a symmetric mesh. On a synthetic structured channel mesh with verified reflection bijection (node-type match 1.0, offset $< 10^{-12}$, edge set invariant), exact mirror-y is admitted and fails on one constructed input (relative L2 1.10). The normalizer control changes the violation only **from 1.1032 to 1.1014**, so learned weights dominate. The boundary is synthetic no-obstacle OOD geometry: the result is not a **calibrated in-distribution magnitude**.

Seeded-fault detection. A 10-mutant catalogue covers boundary-condition, mesh-adjacency, normalization-scale, temporal-rollout, and physical-channel faults. Continuity detects the two boundary-condition faults and the gross normalization fault; symmetry detects one physical-channel and one mesh-adjacency fault; node permutation detects none because the mutants preserve relabeling invariance. Thus 5/10 mutants are detected, with MR-family-specific sensitivity. This is suggestive, not validated localization: detected faults are gross, near-misses remain below threshold, and undetected faults preserve the scored invariants. Rollout accuracy is complementary, but the scope remains one SUT, one checkpoint, and one injected-fault catalogue.

The remaining subsections widen scope: the MGN roster tests denominator robustness, the operator-floor sweep calibrates numerical decidability, and the fault catalogue bounds detector geometry. Cross-family PINN and FNO executions test transfer beyond the original MeshGraphNets workflow.

5.3. Same-task replication

The pilot is replayed on a K=6 MeshGraphNets roster and three official held-out DeepMind trajectories. S0 reuses the pilot checkpoint, so the effective unit is still a within-family roster, not six independent SUTs. Node permutation is exact throughout; mirror-y OOD stress fails on 180/180 cells; the reference-relative conservation diagnostic passes on 162/162 cells while absolute conservation remains deferred; and exact symmetric-mesh mirror-y fails on 18/18 cells. These counts remove the single-trajectory denominator but remain descriptive within one architecture family and dataset.

Same-task multi-architecture replication. Three cylinder-flow executions test implementation dependence. S4/S5 wider/deeper MeshGraphNet variants reproduce the pattern (2/2 node-permutation passes, 60/60 mirror OOD failures, 54/54 conservation-diagnostic passes, 6/6 exact-symmetry failures); a PointMLP coordinate network also reproduces it (9/9, 10/10, 9/9, and 3/3, median rollout relative L2 0.0298); and NVIDIA PhysicsNeMo `MeshGraphNet` reproduces the node-permutation pass and mirror-y OOD-stress failure on official cylinder_flow TFRecords. The unchanged predicate therefore survives implementation changes on one task and dataset, without supporting a cross-dataset rate.

5.4. Second-task gate discrimination: compressible airfoil flow

A direct gate test is whether changed physics changes verdicts. We apply the same predicate to DeepMind **airfoil**, an official CFD dataset with SU2-simulated compressible transonic flow and non-zero angle of attack. A PhysicsNeMo `MeshGraphNet` with the official 2.33M-parameter architecture is trained for 40 epochs on 100 trajectories and evaluated across a K=6 roster on 40 test trajectories (240 cells). The verdict structure changes for physical reasons: node permutation remains admitted and exact (240/240, relative L2 0.0); incompressible $\nabla \cdot u = 0$ is rejected because density varies by a median factor of 2.13x; compressible unsteady mass conservation is the correct replacement but its absolute discrete form is deferred, with a reference-relative median residual ratio of 1.01; and mirror-y chord symmetry is rejected because angle of attack violates the boundary precondition. The high rollout error at this training budget (median relative L2 0.92) is only a training-state diagnostic. This is second-task gate-discrimination evidence, not an official-checkpoint or SOTA airfoil claim.

5.5. Operator-floor resolution sweep

Proposition (C53: P1 operator-floor soundness). For P1 constant-per-cell divergence on shape-regular triangular meshes and C^2 divergence-free fields with Hessian bound M, the local floor is bounded by $C_{\mathrm{shape}}Mh$. An absolute divergence-free MR verdict is admissible only when its tolerance dominates this floor. Limited to this operator class: it gives no arbitrary/degenerate-mesh, non-P1, learned-output, boundary-mismatch, reliability, or fault-detection guarantee.

The P1 sweep over nine symmetric-mesh resolutions gives slope 0.984 with 95% CI $[0.975, 0.992]$ and $\mathrm{R}^2 = 0.9999$, matching O(h) (Figure 3). For the deployed mesh, an analytic-Hessian predictor matches the measured floor within 0.5% (1.337 vs 1.343), and an a-priori Hessian bound dominates it at every resolution. A second unstructured Delaunay topology gives slope 0.983 (95% CI $[0.953, 1.014]$, $\mathrm{R}^2 = 0.999$), with unstructured/structured floor ratio 1.06 median and 1.33 maximum. Thus the gate uses a derived, topology-stable floor for these two mesh families; degenerate meshes, non-P1 operators, boundary mismatch, and discontinuous fields remain outside the theorem. This justifies deferring absolute conservation and reporting only the reference-relative non-regression guard; a flux-form finite-volume operator admits a separate executable conservation verdict (claim C34).

5.6. Coverage implication: fault-detection robustness across the multi-checkpoint roster

The 10-mutant catalogue is replayed against the K=6 roster with five input-permutation seeds (30 trials per mutant per detector) and predeclared severity sweeps for NS_double_scale and PC_zero_vy. Detection uses pilot thresholds: node-permutation tolerance 10^{-5}, conservation ratio 1.5, and mirror-y relative-change 0.5. Wilson intervals are over (SUT, seed) cells. This is the primary MGN stress-test denominator, not the whole fault-evidence inventory.

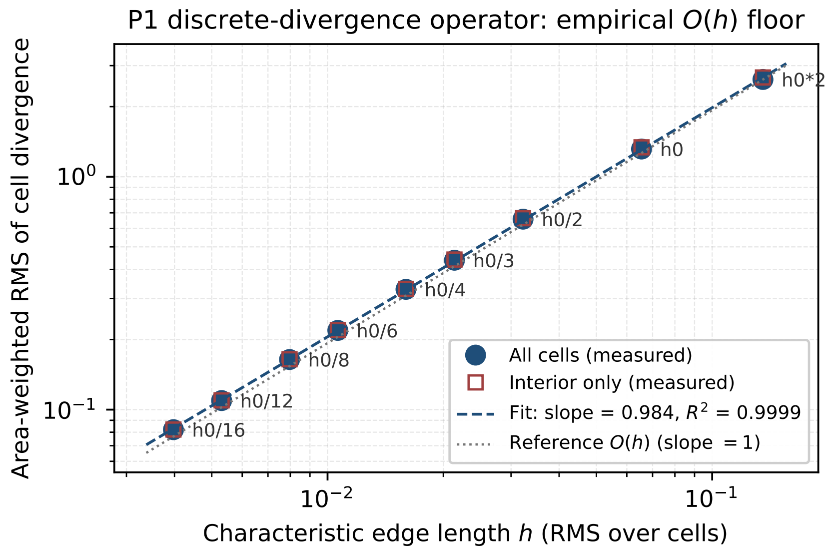

Figure 3: P1 divergence operator-floor calibration.

Table 6 summarizes coverage geometry: each admitted relation measures one invariant family, so blind cells are structural, not sampling noise.

Aggregate reading. The 5/10 union detection rate reproduces across S0–S3 and nearly across S4/S5. Continuity is sensitive to boundary/scale faults; mirror-y is sensitive to physical-channel and adjacency faults; invariant-preserving faults remain invisible. The 60-entry catalogue and two adversarial mutants make the blind region explicit: A1 is detected only by mirror-y, and A2 escapes every detector. On airfoil, mirror-y is excluded by non-zero angle of attack, so mirror-only cylinder-flow coverage disappears. The sibling adds breadth across seven program types and five CPU-only replays.

Independent periodic-advection primary workflow. The **periodic-advection primary workflow** adds an independent synthetic 2D scalar advection task with six deterministic NumPy-trained periodic-convolution SUTs and 10 held-out fields per SUT. It records MR cards, checkpoints, source/follow-up outputs, ledgers, rubric decisions, and verdicts. Periodic translation is admitted and gives **60/60 translation passes**; mass/mean conservation is admitted and gives **60/60 mass-conservation passes**; the **fixed-velocity mirror candidate is rejected** for 6/6 SUTs because mirroring the scalar field without transforming velocity changes the transport

Table 6: Coverage-geometry matrix.

Detector / relation family	**Visible directions**	**Blind or excluded directions**	**Scope and boundary**
Node permutation	Representation contract passes exactly; also admitted on airfoil	Canonical injected faults preserve node relabeling, so no mutant is detected	Sanity/adapter evidence, not fault-coverage evidence
Continuity / conservation	Boundary-condition faults and gross normalization faults; shared continuity $\rightarrow$ normalization-scale signal on airfoil	Multiplicative output scaling sweep remains undetected (0/6, CI $[0.00, 0.39]$)	Measures conservation-like invariants only; absolute MGN conservation remains deferred
Mirror-y symmetry	Physical-channel and mesh-adjacency faults; partial v_y zeroing detected through $p = 0.99$ (6/6)	Exactly uniform $p = 1.0$ is blind (0/6); mirror-y excluded on non-zero-angle airfoil	Coverage disappears when the relation is inadmissible
Admitted suite	5/10 canonical MGN mutants detected; reproduces across S0–S3, with 4/10 on S4/S5	Invariant-preserving faults remain invisible, including adversarial escapes	Bounded stress-test coverage, not real-world defect-detection rate
Cross-family / cross-program checks	PINN/FNO probes and sibling programs reproduce the invariant-based reading	Trained-mutant MGN/PointMLP evidence and closed-form PINN/FNO probes are not interchangeable	Breadth evidence only; no calibrated coverage law

problem. This is full rubric-to-verdict evidence on an independent synthetic PDE task, **not production CFD or real-defect evidence**.

5.7. Interpretation and boundaries

The study's value is relation-level interpretation, not universal fault finding. Retained MRs complement accuracy/UQ diagnostics, control fault attribution, and make assumptions inspectable; they do not license superiority, reliability, geometry-independent rates, validated localization, real-world defect-detection rates, zero-cost MR discovery, or measured person-hour savings. Two overclaims are blocked: asymmetric-mesh mirror-y remains OOD stress rather than a valid-domain fault oracle, and absolute discrete-divergence is deferred rather than recorded as a conservation pass. Symmetric meshes, airfoil, PINN, and FNO show that these verdicts are not only post-hoc labels.

Boundary of claims.

The blocked list remains active: no external-dataset or geometry-independent rates, AeroGraphNet/DoMINO results, baseline superiority, real-world defect rates, validated localization, runtime, reliability, or model-accuracy claims. The supported claim is auditable MR identification and execution for the studied SciML SUTs.

Implications for SciML testing.

The workflow turns implicit expert checks into explicit MR assets that complement residuals, UQ, and accuracy; cross-SUT calibration remains future work.

When to use, and at what cost. The gate is most useful when an OOD surrogate has a physically necessary relation that can be made numerically decidable. It requires domain, numerical, and testing expertise; this paper provides records and runners, not person-hour, agreement, or maintenance-cost measurements.

6. Threats to Validity

We classify threats following Verdecchia et al. [46] and report against the Empirical Standards for software engineering research [47].

Construct validity. The construct is validity-gated MR asset construction, not model correctness. Wrong physics, boundary conditions, mappings, operators, or tolerances could change a verdict, so each executable relation requires an MR card, explicit preconditions, typed verdicts, and a floor argument. Limits remain: the P1 floor is for the deployed symmetric-mesh setting and one Delaunay topology, not arbitrary meshes; the conservation diagnostic is a non-regression guard; and seeded-fault rates do not estimate real-world defect prevalence.

Internal validity. Setup, checkpoints, seeds, preprocessing, normalization, and runtime nondeterminism may affect verdicts. Manifests, ledgers, and raw outputs are recorded; cylinder-flow checks are repeated across K=6 checkpoints, three trajectories, variants, PointMLP, and PhysicsNeMo. These repetitions reduce single-run risk but do not make cells independent samples. Airfoil is read for typed gate discrimination, not model accuracy, because rollout error remains high.

External validity. Evidence spans cylinder flow, airfoil, PINN/FNO Burgers and heat runs, periodic advection, RealPDEBench, a five-unit external issue/PR/commit-linked semantic witness corpus, and sibling cross-program checks. This weakens the self-made-task objection more than a single component witness, but still does not support general reliability, production-CFD validation, representative real-defect rates, broad framework correctness, new MR discovery, or superiority.

Baseline and comparator validity. Rollout accuracy, residual/UQ diagnostics, generic templates, and LLM-assisted candidates are scope contrasts, not defeated competitors. LLM baselines depend on prompts, model versions, and simulated raters; the deterministic rubric, not LLM output, is the validity authority.

Conclusion validity. Counts and intervals are used only where the sampling unit supports them. Many cells share checkpoints, trajectories, or generated cases; the conclusion is bounded empirical utility for RQ4, not calibrated reliability or coverage.

Reproducibility. Old runtimes, public loaders, and non-redistributable inputs limit exact reruns, so the package prioritizes scripts, manifests, logs, ledgers, and available outputs. CPU replay covers floors, validators, typed verdicts, checkpoints, and ledgers; PhysicsNeMo/airfoil training needs GPU access and public DeepMind TFRecords.

7. Future Work

Future work should extend numerical-decidability theory beyond shape-regular P1 triangles; calibrate the domain-violation axis; evaluate production-scale SUTs, datasets, independent real-defect corpora, and independently authored mutants; and automate MR-card authoring, precondition checking, and ledger generation while measuring effort, agreement, and maintenance cost.

8. Conclusion

This paper asked when a physics-derived MR for SciML surrogate software is sound enough to use as an oracle-free test. The answer is a numerical-decidability admissibility criterion: check physical basis, domain preconditions, representation mapping, and operator-floor dominance before executing a relation, then record the result as an MR card, executable asset, ledger entry, and typed verdict.

The evidence is bounded but artifact-backed. Cylinder-flow MeshGraphNets separate a representation pass, mirror-y OOD stress, deferred absolute conservation, and a reference-relative diagnostic; operator-floor analysis explains the deferral; airfoil, PINN, FNO, periodic-advection, and cross-program executions show that the same gate changes or preserves verdicts according to relation conditions; and seeded faults expose admitted-set blind spots without implying general defect-detection rates. The contribution is methodological: physically meaningful SciML metamorphic tests require explicit validity conditions, numerically decidable verdicts, executable assets, raw evidence, and visible claim boundaries.

Preprint

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CRediT authorship contribution statement

Meng Li: Conceptualization, Methodology, Software, Writing – original draft. **Xiaohua Yang:** Supervision, Formal analysis, Writing – review & editing. **Jie Liu:** Investigation, Validation. **Shiyu Yan:** Data curation, Visualization.

Declaration of competing interest

The authors declare no conflict of interest.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used generative AI assistants for code scaffolding, prose copy-editing, and reference cross-checking. After using these tools the authors reviewed and edited the content as needed and take full responsibility for the content of the publication. No generative AI was used to fabricate, alter, or invent experimental data, results, or citations; every numerical claim is backed by a tracked artifact under `research_assets/runs/` and a claim-ledger entry.

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Data availability

The replication package is archived on Zenodo at https://doi.org/10.5281/zenodo.20702952, with source at https://github.com/meng004/Domain-Validity-Gated-MR-for-SciML. It contains the manuscript source, MR cards, validators, claim ledger, run manifests, metric ledgers, and committed derived outputs under research_assets/runs/. Minimum-MR-SubSet audit/rerun artifacts cite commit 9ef862ec37335b4834d0a1fb38b4b613af702f34. DeepMind cylinder-flow and airfoil TFRecords are public benchmark inputs staged by the workflow runners rather than redistributed in the manuscript package. GPU-dependent reruns and credential requirements are documented in REPRODUCIBILITY.md; absent credentials cause fail-closed precondition checks rather than silent substitution.

Supplementary material

The detailed claim-to-evidence map, cross-program breadth notes, secondary baseline audit, cross-family PINN/FNO execution details, RealPDEBench foil preflight witness, and external defect-witness corpus are supplied as supplementary material. The authoritative runtime mapping remains in research_assets/experiments/claim-ledger.yml; the submitted supplement preserves the extracted tables and boundary statements used to audit the manuscript claims.

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